QP CODE: 24900053

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP(HONOURS) REGULAR EXAMINATION NOVEMBER 2024

FIRST SEMESTER

Discipline Specific Core Course-MG1DSCCHE100-FUNDAMENTALS OF CHEMISTRY-I

(2024 ADMISSION ONWARDS)

Duration:**1.5 Hours** Maximum marks: **50**

Part A

Multiple Choice Questions

Answer **all** questions

Each question carries **1** mark

1.Which of the following is not a spectral series in the line spectrum of hydrogen? [K] [1]

c) Hund's

2. The uncertainty principle was developed by [K] [1]

c) Heisenberg

3. Which series of lines in the line spectra of hydrogen atoms lies in the visible region of electromagnetic radiation? [K] [1]

b) Balmer

4. Homolytic fission of chlorine molecule in presence of UV light gives [A] [3, 4]

c) chlorine free radical

5. Organic ions which contain positively charged carbon atom are called [K] [3, 4]

a) Carbocations

6. Which scientist proposed the concept of matter waves? [K] [1]

c) Louis de Broglie

7. Which of the following is an organic compound commonly used as a fertilizer to help plants grow? [K] [2]

a) Urea

8. Which of the following is not the characteristic property of metalloids? [K] [5]

a) They are highly malleable and ductile

9. While moving down in a group, the metallic character: [K] [5]

a) Increases

10. What is the primary characteristic of a covalent bond? [K] [5]

b) Sharing of electron pairs

Part B

Short answer type questions

Answer any **four** questions

Each question carries **3** marks

11. Explain any three limitations of Bohr theory [U] [1]

- Failure in explanation of fine spectrum of hydrogen, Zeeman effect, Stark effect, shape of molecules
- Not in agreement with de Broglie' wave nature of electron and Uncertainty principle

(Give full credit for any three relevant points)

12. Using Heisenberg's uncertainty principle, calculate the uncertainty in velocity of an electron if uncertainty in its position is 10^{-11} m. Given h=6.626 x 10^{-34} kg m^2/s and mass of electron is 9.1 x 10^{-31} kg [A] [1]

$\Delta x \times \Delta p \geq h/4\pi$

$\Delta v = \frac{6.6 \times 10^{-34} \text{ kg m}^2/\text{s}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 10 \times 10^{-11}} = 5.79 \times 10^6$ m/s.

(Equation only - 1 mark, Equation with correct answer-3 marks)

13. Using curved arrow notation, illustrate the formation of reactive intermediates through heterocyclic bond cleavage with a suitable example. [A] [3, 4]

- Definition of heterocyclic bond cleavage with a suitable example
  In this process one of the atoms acquires both the bonding electrons when the bond is broken depending upon electronegativity.
- Transfer of bonding electrons using curved arrow notation
  Eg: Heterolytic cleavage of HCl

(Definition only-1 mark, definition with any relevant example- 3 marks)

14. Why is chloroacetic acid more acidic than acetic acid? [U] [3, 4]

- -I effect of chlorine in case of chloroacetic acid assisting in H^+ ion dissociation

(Give full credit to any relevant explanation)

15. List the merits and demerits of Mendeleev's periodic table. [U] [5]

(Out of syllabus – Give full credit)

16. Interpret why hydrogen occupies a unique position in the Modern Periodic Table [A] [5]

- Similarity with Group 1 and Group 17
  Hydrogen occupies a unique position in the Modern Periodic Table because it shares properties with both alkali metals and halogens, but does not fit perfectly into either group. Like alkali metals, it has one valence electron and can lose it to form H^+, yet it can also gain an electron to form H^-, similar to halogens.
- Formation of ionic and covalent bond

(Give full credit to any relevant explanation)

Part C

Short Essay type questions

Answer any **four** questions

Each question carries **7** marks

17. Define carbocations and describe two methods of preparing them. Discuss the factors that contribute to the stability of a tertiary carbocation compared to primary and secondary carbocations. [A] [3, 4]

- Definition of carbocation
- Preparation (Heterolytic bond fission, Protonation or any relevant method)
- Stability by +I effect and hyperconjugation

(Definition- 1 mark, Preparation- 3 marks, Stability- 3 marks)

18. Examine the common properties and trends of metals, non-metals and metalloids in the periodic table. [A] [5]

Metals include the alkali metal, alkaline earth, transition metal, lanthanide and actinide groups.

- At room temperature, they are usually solids (except mercury).
- They usually have high boiling and melting points.
- They are shiny having a metallic luster.
- They are good conductors of heat and electricity.

Non-metals exhibit a variety of physical and chemical properties that are distinct from those of metals. Nonmetals display the following properties:

- They are brittle and are not malleable or ductile.
- At room temperature, they are usually solids or gases.
- They usually have low boiling and melting points (except for carbon and boron).
- They are poor conductors of heat and electricity.

Metalloids exhibit some properties of metals as well as non- metals. Hence, they are also known as semi-metals. Boron, Silicon, Germanium, Arsenic, Antimony, and Tellurium are the most commonly recognized metalloids.

- They may have a metallic or dull appearance.
- They are usually brittle and fair conductors of electricity.
- They may gain or lose electrons during chemical reactions.
- They form alloys with metals.

(Give full credit if any three relevant properties of each is given with examples)

19. Discuss the atomic line spectrum of hydrogen [U] [1]

- Excitation from ground state to higher energy levels by absorbing energy
- Returns to lower-level emitting radiations.
- These transitions create series like the Lyman, Balmer, Paschen, Brackett, and Pfund, corresponding to electrons falling to the first, second, third, fourth, and fifth energy levels, respectively.

(Explanation- 4 marks, Diagram-3 marks)

20. Explain the different types of hybridisations with examples. [U] [2]

- Types of hybridisation with examples (sp^3 (Methane, Ethane), sp^2 (Ethene), sp (Ethyne) or any relevant examples)
- Detailed explanation including geometry, bond angle.
  sp^3 -tetrahedron (109^0 28')
  sp^2- equilateral triangle (120^0)
  sp- linear (180^0)
- Give ground state and excited state configuration to explain hybridisation

(Explanation- 4 marks, Example for each hybridisation-3 marks)

21. What are quantum numbers? Discuss the significance of quantum numbers. [U] [1]

- Definition of quantum numbers
- Types and significance (Principal quantum number (n), Subsidiary or azimuthal quantum number (l), Magnetic quantum number (m), Spin quantum number (s))
- The principal quantum number (n) determines the energy level and size of an orbital, while the azimuthal quantum number (l) defines its shape. The magnetic quantum number (m) specifies the orientation of the orbital in space, and the spin quantum number (s) indicates the direction of the electron's intrinsic spin, either +½ or -½.
- Values of quantum numbers

  'n' can have values, 1, 2, 3, 4 etc
  'l' and can have values from 0 to n – 1
  The values of m are -'l' through zero to + 'l' and thus there are (2l + 1) values possible
  spin quantum number can have only values +1/2 or -1/2.

(Definition- 2 marks, Significance - 5 marks)

22. Write short notes on covalent and coordinate bonds. [U] [5]

- Definition of covalent bond and coordinate bond
  When both the atoms taking part in chemical combination are short of electrons than the nearest noble gas, the bond is formed by mutual sharing of electrons. This type of bond is called covalent bond.
  The method of combination of two atoms by sharing a pair of electrons, which is contributed by only one of the atoms is termed co- ordinate bond.
- Features of covalent bond and coordinate bond
- Examples with structure

(Explanation of covalent bond with example- 3.5 marks,

explanation of coordinate bond with example- 3.5 marks)